

Quiz 1: What the Judges Actually Score

Review of Session 1. No new material. ~10 minutes.

Section A: Vocabulary

1. State Locard's Exchange Principle.
2. What is the difference between a **class characteristic** and an **individual characteristic**? Give one example of each for a shoe.
3. What is the difference between **direct** and **circumstantial** evidence?
4. True or false: "circumstantial evidence is weak evidence." Explain.
5. What is the difference between a **presumptive** test and a **confirmatory** test? Which kind does a Crime Busters team mostly do?
6. Why does **chain of custody** matter?

Section B: How the test is scored

1. Roughly what percentage of the score is qualitative analysis (powders, liquids, metals)?
2. Roughly what percentage is the written analysis?
3. On a 50-minute test, which section should you protect time for, and why?

Section C: Thinking like a grader

1. A suspect has a strong motive but no physical evidence ties them to the scene. Do you accuse them? What do you write instead?
2. A powder at the scene fizzes with acid *and* leaves undissolved residue. Is that a failed test? What is it telling you?
3. You find a blue cotton fiber at the scene. Two suspects wear blue cotton. Is this a strong match or a common one?

Answer Key

Section A

1. Every contact leaves a trace — when two objects touch, material transfers both ways (the criminal leaves something and takes something away).
2. Class = puts an item in a group (shoe size, tread design, brand). Individual = unique to one source (a specific cut or gouge or wear mark in the sole).
3. Direct proves a fact by itself (eyewitness, video, confession). Circumstantial requires an inference (her hair was there, so she was probably there).
4. False. A stack of circumstantial evidence all pointing the same way is a strong case; most real cases are built this way.
5. Presumptive = fast, sensitive, not specific — tells you what something *could* be.
Confirmatory = identifies it for certain. Crime Busters is almost all presumptive testing plus comparison.
6. So the evidence presented in court is provably the same evidence collected at the scene, untampered — a broken chain makes evidence inadmissible.

Section B

1. ~50%.
2. ~25%.
3. Qualitative analysis — it's half the points and it's slow; also leave time for the analysis writeup (highest points per minute once IDs are done).

Section C

1. No. You note the motive but explain there is no supporting physical evidence, so motive alone isn't enough.
2. Not a failed test — it's a mixture signal, or it's an insoluble carbonate (chalk/lime) rather than baking soda. A partial result is information.
3. A common match — blue cotton is everywhere. It narrows but doesn't point to one person.